

t -tests

CSU, Chico Math 314

2025-10-20

outline

Recap

One Sample

- one sample

- paired data

- paired example

Two Sample

- standard error

- test statistic

- confidence interval

- degrees of freedom

- two sample example

Take Away

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Recap: Inference

Statistical inference revolves around a common theme: assume Y is a random variable such that $E(Y) = \mu$ and $Var(Y) = \sigma^2$. We estimate and/or make statements about μ with

$$\bar{Y} \pm t_{df}^* \cdot s_{\bar{Y}}, \quad \text{and} \quad t_{df} = \frac{\bar{Y} - \mu_0}{s_{\bar{Y}}}.$$

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One Sample t-test

We have already been doing one sample t-tests and confidence intervals.

```
## 95% CI for mu

se <- s/sqrt(n)
t.star <- qt(1-alpha/2, df=n-1)
ME <- se * t.star
c(phat - ME , phat + ME)    # confidence interval (CI)

## test statistic for hypothesis test
t <- (phat - p0)/(se)

## p-value for two sided test,  $H_1: p \neq p_0$ 
2*(1-pt(abs(t), df))
```

Paired Data, definition

Paired data are intimately connected in some way.

Paired Data

Two sets of observations are paired if each observation in one set has a special correspondence or connection with exactly one observation in the other data set.

Paired Data, by example

Are these data paired or not?

- ▶ two websites' prices for the same book
- ▶ left and right eye sight ratings by person
- ▶ upper versus lower bird beak lengths
- ▶ weights of male and female babies

Paired Data, t-test

If the data are paired, their difference has direct and interpretable meaning both in english and in statistics; in mathematical terms, $X_{i,diff} = X_{i,a} - X_{i,b}$ has meaning. Therefore

$$\bar{X}_{diff} \quad \text{and} \quad s_{\bar{X}_{diff}}$$

are simply fancy ways to write new random variables representing this difference.

Paired Data, one sample t-test example

Are textbooks actually cheaper online? Compare the price of textbooks at the University of California, Los Angeles' (UCLA's) bookstore and prices at Amazon.com. Seventy-three UCLA courses were randomly sampled in Spring 2010.

```
books <- read.csv("https://raw.githubusercontent.com/  
njlytal/MATH314/main/books.csv")  
## look at data in RStudio  
## what plot should we make?
```

Paired Data, one sample t-test example

Plot the data!

```
suppressMessages(library(ggplot2))  
ggplot(data = books, aes(uclaNew - amazNew)) +  
  geom_boxplot()  
  
or  
  
ggplot(data=books, aes(uclaNew, amazNew)) +  
  geom_point() + geom_abline(intercept=0, slope=1)
```

Here, we can obtain a boxplot of the differences in each book's price, or a scatterplot where each dot represents a book and its two prices on the two axes.

Paired Data, one sample t-test example

Calculate and interpret a 95% confidence interval of the difference in Amazon.com versus UCLA's book prices.

Paired Data, one sample t-test example

```
## Define variables
diff <- books$uclaNew-books$amazNew # differences
Xbar <- mean(diff) # sample statistic
n <- length(diff) # sample size
df <- n-1 # degrees of freedom

# Define margin of error
t.star <- qt(1-alpha/2,n-1) # critical value
se <- sd(diff)/sqrt(n) # sample error
ME <- t.star*se # margin of error

## Define confidence interval
CI <- c(Xbar - ME, Xbar + ME)
CI

## [1] 9.435636 16.087652
```

Paired Data, one sample t-test example

We are 95% confident that UCLA's new book prices are greater than Amazon.com's books by between \$9.44 and \$16.09.

Paired Data, one sample t-test example

Set up, evaluate, and conclude in context a hypothesis test at $\alpha = 0.05$.

Paired Data, one sample t-test example

The natural hypotheses are

$$H_0 : \mu_{diff} = 0 \text{ versus } H_1 : \mu_{diff} \neq 0.$$

```
t <- (mean(diff) - 0) / (sd(diff)/sqrt(n)) # test stat
2*(1-pt(abs(t), n-1)) # p-value

## [1] 6.92757e-11
```


Paired Data, one sample t-test example

Because $p\text{-value} < 0.0001 < \alpha = 0.05$, we reject H_0 . There is insufficient evidence to claim that Amazon.com and UCLA's book prices are the same.

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Two Sample t-test

Two sample t-tests estimate the difference between two population means from two independent samples of data. We estimate $\mu_a - \mu_b$ with the point estimator $\bar{X}_a - \bar{X}_b$.

Two Sample t-test, standard error

We estimate the variance of the point estimator $\bar{X}_a - \bar{X}_b$, namely $Var(\bar{X}_a - \bar{X}_b) = \sigma_{\bar{X}_a - \bar{X}_b}^2$, with

$$s_{\bar{X}_a - \bar{X}_b}^2 = \frac{s_a^2}{n_a} + \frac{s_b^2}{n_b}.$$

Two Sample t-test, test statistics

$$t_{df} = \frac{(\bar{X}_a - \bar{X}_b) - (\mu_a - \mu_b)}{\sqrt{s_{\bar{X}_a - \bar{X}_b}^2}}$$

Two Sample t-test, confidence interval

$$(\bar{X}_a - \bar{X}_b) \pm t_{df}^* \sqrt{s_{\bar{X}_a - \bar{X}_b}^2}$$

Two Sample t-test, degrees of freedom

It's common to use a conservative approximation for the degrees of freedom for a t-test of the difference of two independent means,

$$df = \min(n_a - 1, n_b - 1).$$

Two Sample t-test, example

Consider the data set `ape::carnivora`. Calculate a 98% confidence interval for the difference in mean longevity between the two superfamilies¹ Caniformia and Feliformia.

```
suppressMessages(library(ape))  
data(carnivora)
```

¹Technically, these are suborders, but they are listed in this data set as superfamilies

Tangent on NAs and R functions

Some helpful functions to help you avoid NAs in R are

```
anyNA()  
is.na()  
na.omit()
```

Additionally, the `dplyr` package has the "pipe operator" (`% > %`) which lets you chain two functions together. Whatever is to the left of `% > %` will be input as the **first** argument of the function on the right.

```
# UCLA book prices are put as 1st argument of mean()  
books$uclaNew %>% mean()  
  
[1] 72.22192
```

Two Sample t-test, example

A 98% CI, difference in longevity between Caniformia and Feliformia.

```
(d <- carnivora %>%  
  select(SuperFamily, LY) %>% # only two columns  
  na.omit() %>% # be careful with placement  
  group_by(SuperFamily) %>% # mean/var by SuperFamily  
  summarise(n=n(), v=var(LY), xbar=mean(LY)))
```

A tibble: 2 x 4

##	SuperFamily	n	v	xbar
##	<fct>	<int>	<dbl>	<dbl>
## 1	Caniformia	24	7139.	192.
## 2	Feliformia	25	2576.	172.

This "chain" extracts the variables of interest, isolating the data down to the columns we want and summarizing them for each group.

Two Sample t-test, example

A 98% CI, difference in longevity by Caniformia and Feliformia.

```
alpha <- 0.02
xbarC <- 192.4583; xbarF <- 171.96
diff <- (xbarC - xbarF)  # difference of means
nC <- 24; nF <- 25
vC <- 7139.216; vF <- 2576.04
t.star <- qt(1-alpha/2, df=min(c(nC, nF)-1))
se <- sqrt(vC/nC + vF/nF)
ME <- t.star*se

c(diff - ME, diff + ME)

## [1] -29.53083  70.52743
```

These variables are defined from the output of the previous slide's code.

Two Sample t-test, example

We are 98% confident that the population difference in mean longevity between the superfamilies Caniformia and Feliformia is between -29.53 and 70.53 months.

Two Sample t-test, example

Set up, evaluate, and conclude in context a hypothesis test at $\alpha = 0.02$.

Two Sample t-test, example

The natural hypotheses are

$$H_0 : \mu_C = \mu_F \text{ versus } H_1 : \mu_C \neq \mu_F.$$

```
# test statistic  
t <- ((xbarC - xbarF) - 0) / sqrt(vC/nC + vF/nF)  
2*(1-pt(abs(t), min(c(nC, nF)-1))) # p-value  
  
## [1] 0.3163636
```

NOTE: You can equivalently say $H_0 : \mu_C - \mu_F = 0$

Two Sample t-test, example

Because $p\text{-value} = 0.32 > \alpha = 0.02$, we fail to reject H_0 . There is insufficient evidence to claim that the true difference in mean longevity between Caniformia and Feliformia is different.

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Whatever form the problem takes, we're doing the same steps: find parameters of interest, insert point estimates, and calculate the standard error of the estimators.

- ▶ Paired data
 - ▶ Two variables are intimately connected $\Rightarrow$ their difference has meaning
 - ▶ create one variable from the two $\Rightarrow$ one sample t-test
- ▶ Two sample data
 - ▶ Two variables are independent
 - ▶ Point estimate is difference of means
 - ▶ Standard error follows from independence
- ▶ CLT via standardization makes this all possible.

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Adapted from notes by Dr. Roualdes.